



Summary Report for Nokia

Leveraging Machine Learning to Optimize Fault Assignment and Resolution

Project Title: **Predicting the Responsible Team & Groups for Software Fault Resolution**

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Executive Summary:

Nokia's ability to maintain high operational efficiency and customer satisfaction depends heavily on how quickly and accurately support tickets are resolved. Currently, assigning each new software fault ticket to the appropriate group is a manual task. This process is time-consuming, prone to error, and places a significant burden on support and development teams.

In collaboration with Tel Aviv University, we developed a machine learning-based solution that predicts the responsible group for a ticket at the moment it is created. The model shows strong predictive performance and has the potential to reduce delays, minimize misassignments, and increase both engineering productivity and service quality. It is ready for pilot deployment and can be integrated with minimal changes to Nokia's existing systems.

The Business Challenge:

- Approximately 10 new software issues reported each day
- Up to 30 days average resolution time per ticket
- Manual triage requires significant resources and delays resolution
- Developers productivity is impacted by misrouted or poorly assigned tickets
- Delays and inefficiencies can lead to reduced customer satisfaction and reputational risk

Research Question/Project Goal:



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Faculty of Engineering
Tel Aviv University



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Can machine learning optimize issue resolution times in telecommunications?

Solution Approach:

We analyzed a dataset of over 10,000 historical fault tickets from Nokia's Jira system. After cleaning and filtering to retain only relevant cases, we engineered features from structured metadata and used GPT-based embeddings to extract meaning from free-text descriptions. Based on these features, we developed a custom feedforward neural network and used the Optuna framework to optimize its performance. The model was evaluated using multiple metrics including accuracy, F1 score, and AUC-ROC to ensure a balanced view of its predictive ability.

Key Results:

- Validation Accuracy: 61.8%
- F1 Score : 0.61
- AUC-ROC: 0.905
- Precision:0.6387
- Recall : 0.6012
- Weighted Averages:
 - Precision: 0.6167
 - Recall: 0.6179
 - F1 Score: 0.6150
- High-performing groups (F1 > 0.85): CNF, UI, SS7
- Underperforming groups: Support, SPV (due to class imbalance and fewer examples)

Performance Insights:

The model achieved a validation accuracy of 61.8% and an AUC ROC of 0.905, indicating strong overall performance in distinguishing between ticket categories. The macro average F1 score was 0.61, with precision and recall at 0.64 and 0.60 respectively. Performance was particularly robust in high



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volume areas such as CNF, UI, and SS7, where F1 scores exceeded 0.85. In contrast, weaker results emerged in smaller or less represented groups such as Support and SPV, primarily due to limited training examples and class imbalance. To enhance predictions for these groups, further data collection is necessary, along with a strategic decision regarding future actions, which could involve investing in additional data gathering or integrating alternative workflows. Such measures will help ensure the model's continuous improvement and consistent performance across all ticket categories.

Operational Benefits:

The implementation of this predictive model significantly reduces the manual workload involved in fault ticket triaging by support engineers. By automating the immediate assignment of tickets to the appropriate teams upon their creation, the model notably improves operational efficiency. Specifically, the automated system enhances the precision and speed of team allocation processes, effectively minimizing delays and substantially accelerating the resolution timeline for software-related issues. As a result, engineering teams are freed from repetitive ticket management tasks, enabling them to allocate their focus to higher-level, strategic responsibilities. This shift notably boosts service reliability and ensures timely delivery, directly enhancing overall customer satisfaction.

Strategic Recommendations:

1. **Initiate Pilot Deployment:** Integrate the model into the live support system for a subset of tickets - Monitor prediction quality and measure reduction in resolution time
2. **Build a Feedback Loop:**
 - o Continuously retrain the model with new data to improve accuracy
 - o Use resolution outcomes to refine predictions and team mappings
3. **Refine Granularity:**
 - o Move from group-level to team-level predictions as more data is collected
 - o Use clustering and smart grouping to address class imbalance



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4. **Ensure Fairness and Explainability:**

- o Regularly evaluate model performance across all groups
- o Implement explainable AI tools like SHAP to increase transparency and trust

Ethical and Technical Considerations:

All personally identifiable information (PII) was excluded or anonymized in the dataset - Model fairness was assessed across different groups, with plans to improve support for underrepresented classes - Neural network architecture limits interpretability, but future iterations will include transparency mechanisms

Conclusions and Future Directions:

The collaboration between Nokia and Tel Aviv University has demonstrated the practical value of applying machine learning to software fault resolution. Together, we developed a predictive model that determines the appropriate support group for each ticket at the moment of issue creation, using structured data and GPT-based text embeddings. The model achieved strong performance across key metrics, including accuracy, precision, recall, and F1 score.

Given these results, we recommend a phased rollout, starting with integration into selected workflows. During this initial stage, Nokia should closely monitor outcomes and gather user feedback to guide future improvements. Future development should focus on improving predictions for underrepresented groups, enhancing transparency with explainable AI methods, and expanding the dataset to reduce performance gaps and bias.

With continued refinement and thoughtful deployment, this system can reduce manual workload, shorten resolution times, and position Nokia as a leader in intelligent support solutions within the telecommunications industry.